



Experiment-7

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1. Aim/Overview of the practical:

To identify the candidate key(s), prime and non-prime attributes, and the highest normal form satisfied by a given relation schema and its set of functional dependencies (FDs), and to decompose relations that are not in a higher normal form so as to remove redundancy.

2. Software/Tools Required:

- Pen and paper / manual FD-closure analysis
- MySQL Workbench (for optional verification of the decomposed schema)

3. Theory:

A) Functional Dependency (FD)

A functional dependency $X \rightarrow Y$ between two attribute sets of a relation means that the values of X uniquely determine the values of Y – i.e. no two tuples can agree on X while disagreeing on Y.

B) Candidate Key

A candidate key is a minimal set of attributes whose closure (computed under the given FDs) covers every attribute of the relation. A key is found by starting from the attributes that never appear on the right-hand side of any FD (these are compulsorily part of every key) and then adding just enough additional attributes until the closure equals the full attribute set.

C) Prime and Non-prime Attributes

An attribute is called prime if it takes part in at least one candidate key of the relation; every other attribute is non-prime.

D) Normal Forms used in this experiment

Normal Form	Condition
1NF	All attribute values are atomic (no repeating groups).
2NF	Relation is in 1NF and no non-prime attribute depends on a proper subset of any candidate key (no partial dependency).
3NF	Relation is in 2NF and for every non-trivial FD $X \rightarrow Y$, either X is a superkey or every attribute in Y is prime (no transitive dependency of a non-prime attribute through another non-prime attribute).
BCNF	Relation is in 3NF and for every non-trivial FD $X \rightarrow Y$, X must be a superkey (stricter than 3NF – no exception for prime attributes).

A relation is placed in the highest normal form whose condition it satisfies, after also satisfying all the lower ones.

4. Practical Set A – Candidate Key & Highest Normal Form

Q1. Relation R(A,B,C,D,E)

Functional Dependencies given:

$$CE \rightarrow D$$

$$D \rightarrow B$$

$$C \rightarrow A$$

Step 1 – Attribute closure / essential attributes

Attributes C and E never occur on the right-hand side of any FD, so both must belong to every candidate key.

$$(CE)^+ = \{C,E\} \rightarrow \text{add } D \ (CE \rightarrow D) \rightarrow \text{add } B \ (D \rightarrow B) \rightarrow \text{add } A \ (C \rightarrow A) = \{A,B,C,D,E\} = \text{all attributes.}$$

Since CE alone already reaches all attributes, CE is the (unique) candidate key.

Step 2 – Candidate Key(s)

$$CK = \{ CE \}$$

Step 3 – Prime / Non-prime attributes

Prime attributes: C, E

Non-prime attributes: A, B, D

Step 4 – Highest Normal Form

Candidate key CE has size 2. The FD $C \rightarrow A$ has C as a proper subset of the key CE, and A is a non-prime attribute – this is a partial dependency, so 2NF is violated. Hence the relation cannot go beyond 1NF.

Highest Normal Form = 1NF

Q2. Relation R(A,B,C,D,E,F)

Functional Dependencies given:

$$AB \rightarrow C$$

$$DC \rightarrow AE$$

$$E \rightarrow F$$

Step 1 – Attribute closure / essential attributes

B and D never occur on the right-hand side of any FD, so both are compulsory in every key, but $(BD)^+ = \{B,D\}$ only – not sufficient alone.

$$(ABD)^+: \text{add } C \ (AB \rightarrow C) \rightarrow \text{add } A,E \ (DC \rightarrow AE, \text{ already have } A) \rightarrow \text{add } F \ (E \rightarrow F) = \text{all attributes. } ABD \text{ is minimal.}$$

$$(BCD)^+: DC \rightarrow AE \text{ fires directly (D,C present)} \rightarrow \text{add } A,E \rightarrow AB \rightarrow C \text{ already satisfied} \rightarrow E \rightarrow F \text{ adds } F = \text{all attributes. } BCD \text{ is also minimal.}$$

Step 2 – Candidate Key(s)

$$CK = \{ ABD, BCD \}$$

Step 3 – Prime / Non-prime attributes

Prime attributes: A, B, C, D

Non-prime attributes: E, F

Step 4 – Highest Normal Form

Take key BCD: the proper subset CD is not itself a key, yet $(CD)^+ = \{A,C,D,E,F\}$, so CD determines the non-prime attributes E and F – a partial dependency. Hence 2NF fails and the relation stays in 1NF.

Highest Normal Form = 1NF

Q3. Relation R(A,B,C,D,E,F,G,H,I)

Functional Dependencies given:

AB $\rightarrow$ C
BD $\rightarrow$ EF
AD $\rightarrow$ GH
A $\rightarrow$ I

Step 1 – Attribute closure / essential attributes

A, B, D never appear on the right-hand side of any FD, so all three must be present in every key.

(ABD)⁺: AB $\rightarrow$ C adds C; BD $\rightarrow$ EF adds E,F; AD $\rightarrow$ GH adds G,H; A $\rightarrow$ I adds I $\rightarrow$ all 9 attributes reached.

No 2-attribute subset of {A,B,D} reaches full closure, so ABD is the unique minimal key.

Step 2 – Candidate Key(s)

CK = { ABD }

Step 3 – Prime / Non-prime attributes

Prime attributes: A, B, D

Non-prime attributes: C, E, F, G, H, I

Step 4 – Highest Normal Form

AB is a proper subset of the key ABD, and AB $\rightarrow$ C makes C (non-prime) dependent on only part of the key – a partial dependency. 2NF is therefore violated, so the relation is only in 1NF.

Highest Normal Form = 1NF

Q4. Relation R(A,B,C,D,E)

Functional Dependencies given:

AB $\rightarrow$ CD
D $\rightarrow$ A
BC $\rightarrow$ DE

Step 1 – Attribute closure / essential attributes

B never appears on the right-hand side of any FD, so B belongs to every key, but B alone is not sufficient.

(AB)⁺ = {A,B,C,D,E} (AB $\rightarrow$ CD, then BC $\rightarrow$ DE) – full closure.

(BC)⁺ = {A,B,C,D,E} (BC $\rightarrow$ DE, then D $\rightarrow$ A) – full closure.

(BD)⁺ = {A,B,C,D,E} (D $\rightarrow$ A, then AB $\rightarrow$ CD, then BC $\rightarrow$ DE) – full closure. All three pairs are minimal.

Step 2 – Candidate Key(s)

CK = { AB, BC, BD }

Step 3 – Prime / Non-prime attributes

Prime attributes: A, B, C, D

Non-prime attributes: E

Step 4 – Highest Normal Form

Only E is non-prime, and it is never derived from a proper subset of a key alone, so 2NF holds. For 3NF, every FD either has a superkey on the left (AB, BC are keys) or a prime attribute on the right (D $\rightarrow$ A, A is prime) – so 3NF holds. For BCNF every determinant must be a superkey; but D alone is not a superkey (D⁺ = {A,D} $\neq$ all attributes) even though it determines only the prime attribute A, so BCNF is violated. Highest normal form reached is therefore 3NF.

Highest Normal Form = 3NF

Q5. Relation R(A,B,C,D,E)

Functional Dependencies given:

BC $\rightarrow$ ADE
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$D \rightarrow B$

Step 1 – Attribute closure / essential attributes

C never appears on the right-hand side of any FD, so C is compulsory in every key, but C alone is insufficient.

$(BC)^+ = \{A, B, C, D, E\}$ directly via $BC \rightarrow ADE$ – full closure.

$(CD)^+$: $D \rightarrow B$ adds $B \rightarrow BC \rightarrow ADE$ now fires (B,C present) adding A,E – full closure. Both pairs are minimal keys.

Step 2 – Candidate Key(s)

CK = { BC, CD }

Step 3 – Prime / Non-prime attributes

Prime attributes: B, C, D

Non-prime attributes: A, E

Step 4 – Highest Normal Form

A and E (the only non-prime attributes) are never produced from a lone proper subset of a key, so 2NF holds. Checking 3NF: $BC \rightarrow ADE$ has a superkey on the left (OK); $D \rightarrow B$ has D which is not a superkey ($D^+ = \{B, D\}$), but B is a prime attribute, so the 3NF exception applies and 3NF still holds. For BCNF, $D \rightarrow B$ requires D itself to be a superkey, which it is not – so BCNF fails. Highest normal form reached is 3NF.

Highest Normal Form = 3NF

Q6. Relation R(A,B,C)

Functional Dependencies given:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

Step 1 – Attribute closure / essential attributes

A^+ : $A \rightarrow B$ adds B, $B \rightarrow C$ adds C = $\{A, B, C\}$ – full closure, so A alone is a key.

B^+ : $B \rightarrow C$ adds C, $C \rightarrow A$ adds A = $\{A, B, C\}$ – full closure, so B alone is a key.

C^+ : $C \rightarrow A$ adds A, $A \rightarrow B$ adds B = $\{A, B, C\}$ – full closure, so C alone is a key.

Step 2 – Candidate Key(s)

CK = { A }, { B }, { C } (each single attribute is its own candidate key)

Step 3 – Prime / Non-prime attributes

Prime attributes: A, B, C (all attributes are prime)

Non-prime attributes: None

Step 4 – Highest Normal Form

Since every attribute is part of some candidate key, there are no non-prime attributes at all, so 2NF and 3NF are automatically satisfied. For BCNF, every FD's left-hand side (A, B and C individually) is itself a candidate key/superkey, so BCNF is fully satisfied – this cyclic relation is already in BCNF, the highest normal form.

Highest Normal Form = BCNF

5. Practical Set B – Normal Form, Candidate Key & Decomposition

Note: In question 1 below, the FD $H \rightarrow J$ introduces attribute J, so the relation is treated as R(A,B,C,D,E,F,G,H,I,J) even though only nine letters were listed in the relation name (a minor typographic slip in the original question).

Q1. Relation R(A,B,C,D,E,F,G,H,I,J)

Functional Dependencies given:

$AB \rightarrow C$

$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

Step 1 – Attribute closure / essential attributes

A, B, D never occur on the right-hand side of any FD, so all three are compulsory in every key.

(ABD)⁺: $AB \rightarrow C$ adds C; $AD \rightarrow GH$ adds G,H; $BD \rightarrow EF$ adds E,F; $A \rightarrow I$ adds I; now H is present so $H \rightarrow J$ adds J – all 10 attributes reached.

No pair among {A,B,D} alone reaches full closure, so ABD is the unique minimal key.

Step 2 – Candidate Key(s)

CK = { ABD }

Step 3 – Prime / Non-prime attributes

Prime attributes: A, B, D

Non-prime attributes: C, E, F, G, H, I, J

Step 4 – Highest Normal Form

AB is a proper subset of the key ABD, and $AB \rightarrow C$ makes the non-prime attribute C depend on only part of the key – a partial dependency, so 2NF fails and the relation is only in 1NF.

Highest Normal Form = 1NF

Step 5 – Decomposition

Applying the 3NF-synthesis decomposition (one relation per FD, using a minimal cover) removes the redundancy caused by the partial and transitive dependencies:

R1 (A, B, C) – from $AB \rightarrow C$

R2 (A, D, G, H) – from $AD \rightarrow GH$

R3 (B, D, E, F) – from $BD \rightarrow EF$

R4 (A, I) – from $A \rightarrow I$

R5 (H, J) – from $H \rightarrow J$

None of R1–R5 contains the full candidate key {A,B,D} together, so one more relation is added to guarantee a lossless-join decomposition:

R6 (A, B, D) – the candidate-key relation

Total new tables formed after decomposition = 6 (R1 to R6), each of which is in 3NF/BCNF and the decomposition is both lossless and dependency-preserving.

Q2. Relation R(A,B,C,D,E)

Functional Dependencies given:

$A \rightarrow B$

$B \rightarrow E$

$C \rightarrow D$

Step 1 – Attribute closure / essential attributes

A and C never occur on the right-hand side of any FD, so both are compulsory in every key.

(AC)⁺: $A \rightarrow B$ adds B; $B \rightarrow E$ adds E; $C \rightarrow D$ adds D = {A,B,C,D,E} – full closure.

A alone gives {A,B,E} and C alone gives {C,D}, neither is sufficient, so AC is the unique minimal key.

Step 2 – Candidate Key(s)

CK = { AC }

Step 3 – Prime / Non-prime attributes

Prime attributes: A, C

Non-prime attributes: B, D, E

Step 4 – Highest Normal Form

A is a proper subset of key AC and $A \rightarrow B$ makes the non-prime attribute B depend on only part of the key; similarly $C \rightarrow D$ makes non-prime D depend on only part of the key. Both are partial dependencies, so 2NF is violated and the relation remains in 1NF.

Highest Normal Form = 1NF

Step 5 – Decomposition

Decomposing on each FD (already a minimal cover, one relation per determinant):

R1 (A, B) – from $A \rightarrow B$

R2 (B, E) – from $B \rightarrow E$

R3 (C, D) – from $C \rightarrow D$

None of R1–R3 contains the full candidate key {A,C}, so a key relation is added:

R4 (A, C) – the candidate-key relation

Total new tables formed after decomposition = 4 (R1 to R4), each now in BCNF, and the join of all four reconstructs R losslessly.

6. Practical Set C – Prime/Non-prime Attributes

Q. Relation P(F1,F2,F3,F4,F5)

Functional Dependencies given:

$F1 \rightarrow F3$

$F2 \rightarrow F4$

$(F1, F2) \rightarrow F5$

Step 1 – Attribute closure / essential attributes

F1 and F2 never occur on the right-hand side of any FD, so both are compulsory in every key.

$(F1F2)^+$: $F1 \rightarrow F3$ adds F3; $F2 \rightarrow F4$ adds F4; $(F1, F2) \rightarrow F5$ adds F5 = {F1, F2, F3, F4, F5} – full closure.

F1 alone gives {F1, F3} and F2 alone gives {F2, F4}, neither is sufficient, so F1F2 is the unique minimal key.

Step 2 – Candidate Key(s)

CK = { F1F2 }

Step 3 – Prime / Non-prime attributes

Prime attributes: F1, F2

Non-prime attributes: F3, F4, F5

Step 4 – Highest Normal Form

F1 is a proper subset of the key (F1, F2), and $F1 \rightarrow F3$ makes the non-prime attribute F3 depend on only part of the key. Likewise $F2 \rightarrow F4$ makes non-prime F4 depend on only part of the key. Both are partial dependencies, so relation P violates 2NF and is only in 1NF. (The dependency $(F1, F2) \rightarrow F5$ is a full dependency on the whole key, so it does not by itself cause a violation.)

Highest Normal Form = 1NF

7. Result

For each of the nine relations given in the practice sets, the candidate key(s) were determined using attribute-closure analysis, the prime and non-prime attributes were identified, and the highest normal form satisfied by the relation was established with proper justification. Wherever a relation was found to violate 2NF, a lossless-join, dependency-

preserving decomposition was carried out using FD-based synthesis, and the resulting number of new tables was reported.

8. Learning Outcomes

- Learned how to compute the closure of an attribute set under a given set of functional dependencies.
- Learned a systematic method to identify all candidate keys of a relation, including cases with multiple candidate keys.
- Understood how to classify attributes as prime or non-prime based on candidate-key membership.
- Learned to check a relation against the conditions for 1NF, 2NF, 3NF and BCNF and identify the highest normal form it satisfies.
- Understood the difference between a partial dependency (violates 2NF) and a transitive dependency (violates 3NF).
- Practiced FD-based (synthesis) decomposition to obtain a lossless-join, dependency-preserving set of higher normal form relations.
- Understood why BCNF is stricter than 3NF, and why some relations can only reach 3NF and not BCNF.